

not happen. Increasingly, information retrieval systems are adopting the use of natural language processing to improve their performance. In the next few columns, we will focus on natural language processing (NLP) and how it is useful in information retrieval systems.

NLP can be applied to the IR tasks of query processing, document matching and relevance ranking. Recall that document matching is done using our main data structures of the dictionary and inverted index. We had discussed using either single word tokens or multi-word tokens from the documents as terms representing the document, and we used these terms to represent the document in the inverted index. While single word tokens (unigrams) can be used directly, we had also considered bi-grams (two consecutive tokens) and trigrams (three consecutive tokens) to improve the matching between query and document. For example, if we had used only unigrams, then if the user searched for 'New York', the retrieved document set could include documents containing just 'New' and just 'York', and we would have to impose a post-processing step to filter out those documents where only one of the two unigrams are present, instead of both. On the other hand, a bi-gram representation of the term 'New York' in an inverted index would have allowed us to directly return only those documents which contained 'New York'. Note that bi-grams need to be maintained in addition to unigrams, since unigrams would still be needed to answer a query for 'York'. Note that a user query for 'Who is the new priest for the town of York?' should not return documents which contain the term 'New York'. Given that many of our information retrieval systems are based on the 'bag of words' model, which does not consider the order of words in the document or query, most of our IR systems would fail this test. On the other hand, consider an NLP analysis of the query which determines that the word 'new' in the query is an adjective to the noun 'priest' and is not part of the noun for 'York'; then the query can be processed to look for 'York' as a unigram instead of retrieving documents which contain both the words 'new' and 'York' without understanding the syntactic structure of the query. Query processing can be enhanced by NLP to consider the syntactic structure, and each word/term in the query can be tagged with additional metadata containing information about its grammatical structure as a noun, verb, etc.

The challenge is to figure out which bi-grams are useful to be represented in the dictionary and which are not. Maintaining all bi-grams (or all trigrams) consumes computation resources and memory. So IR systems typically only maintain a sub-set of all

possible bi-grams over the dictionary. The set of bi-grams that are maintained in the dictionary are chosen depending on how frequently they occur in the document collection without understanding their syntax. Let us consider the query: 'buy prefabricated units.' Users may be looking for a document which contains information on where to buy prefabricated units. Now consider a document which contains the sentence, 'In the forest, they prefabricated units and put them up for temporary use.' In the user query, the bi-gram 'prefabricated units' is a noun phrase whereas in the document, the word 'prefabricated' is a verb. Without using linguistic techniques for syntax parsing, it is not possible to distinguish that the document containing the sentence with the bi-gram 'prefabricated units' is not relevant to the user query. On the other hand, it is possible to use a simple 'Part of Speech' (POS) tagger to tag the various terms in the query, and document and use the POS tags information to aid the matching process. Use of such linguistic techniques in query processing and document matching is made possible by means of NLP.

One of the important points to note is that users typically use nouns very frequently in queries. Therefore, identifying nouns in documents and queries can be quite useful in document retrieval. A well-known NLP technique, known as 'Named Entity Recognition', can be quite useful for this purpose. We will discuss how Named Entity Recognition is possible using natural language processing of text in our next column.

So far, we have been discussing examples of how NLP can be used to aid information retrieval. There are different levels in which NLP can be used for IR. At the morphological level, linguistic techniques can be used in tokenisation, for stemming. At the syntax level, NLP can be used to determine whether a term in a query is a noun or verb, using that information to improve matching. At the semantic level, NLP can be useful in identifying the intended meaning of a word based on the context to improve the matching process. Many of the documents are semi-structured. At the discourse level, it is possible to use the structure of the document to decide whether it is a fact or an opinion that is being reported. For instance, a scientific abstract could start by stating the facts and then conclude by stating the hypothesis it is proposing in the paper. By using the structure of the document to identify a paragraph as an abstract in the document, we can use it as a summary for the document also. At the pragmatic level, word knowledge which lies outside the document collection can be used to aid in query processing and document matching. For example, consider the sentence, 'I went to the bank and withdrew money.' Now, the